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Mamba-Lip: Revolutionizing Lip Reading Generation with Selective State Space Models

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***Abstract*—Lip reading generation, which aims to synthesize realistic facial landmark sequences from textual input, represents a crucial task in visual language processing. While existing methods predominantly rely on Transformer or RNN architectures, they often struggle with capturing long-range dependencies in sequential data and face efficiency challenges when processing lengthy sequences. In this paper, we propose Mamba-Lip, a novel framework that revolutionizes lip reading generation through selective state space models (SSMs). Our approach features two key components: (1) Mamba-Enc, a text encoder that leverages selective SSMs to effectively capture linguistic structures and long-range contextual dependencies; (2) Mamba-Dec, an autoregressive decoder that generates fine-grained facial landmark sequences through causal state space modeling combined with cross-modal attention. The selective mechanism enables our model to dynamically focus on relevant linguistic contexts while efficiently processing long sequences. Extensive experiments demonstrate that Mamba-Lip achieves state-of-the-art performance in both generation accuracy and temporal coherence, while exhibiting superior efficiency compared to Transformer-based counterparts.**

Keywords-Lip Reading Generation, Selective State Space Models, Mamba, Cross-modal Synthesis.

I. INTRODUCTION

Visual language processing has emerged as a critical research domain with applications spanning virtual avatars, assistive technologies, and human-computer interaction. Among these applications, lip reading generation—the task of synthesizing corresponding facial movements from textual input—represents a significant challenge at the intersection of natural language processing and computer vision. This technology holds immense potential for enhancing communication accessibility, creating realistic digital humans, and developing advanced human-machine interfaces.

The evolution of lip reading generation has followed two primary trajectories: audio-driven methods and landmark-based approaches. Early audio-driven methods like Wav2Lip [1] and SyncTalkFace [2] leveraged speech signals to generate synchronized lip movements, while landmark-based techniques such as MakeItTalk [3] utilized facial landmarks as intermediate representations for video synthesis. These methods have progressively improved in generating realistic lip movements, yet they predominantly build upon Transformer or recurrent architectures that face fundamental limitations in handling long sequential dependencies efficiently.

Transformers, while powerful in capturing global dependencies through self-attention mechanisms, suffer from quadratic computational complexity that restricts their applicability to long sequences—a critical limitation given that natural speech often involves extended utterances. Concurrently, RNN-based approaches struggle with capturing distant temporal relationships due to vanishing gradient problems, often resulting in generated sequences that lack long-term coherence. These architectural constraints have impeded progress toward generating seamless, natural-looking lip movements for extended speech segments.

Recent breakthroughs in state space models (SSMs), particularly the Mamba architecture [4], offer a promising alternative for sequential data processing. Unlike traditional SSMs that employ time-invariant parameters, Mamba introduces a selective mechanism that enables content-based reasoning, allowing the model to dynamically adjust information flow based on input context. This property, combined with linear computational complexity, makes SSMs ideally suited for capturing the nuanced relationships between linguistic content and visual articulation in lip reading generation.

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In this paper, we present **Mamba-Lip**, a novel framework that leverages selective state space models to address the fundamental challenges in lip reading generation. Our work represents a significant departure from conventional approaches by replacing the traditional Transformer-based encoder-decoder architecture with a Mamba-based design that efficiently captures both linguistic semantics and visual dynamics. Our main contributions are summarized as follows:

- We introduce Mamba-Enc, a text encoder based on selective SSMs that effectively captures hierarchical linguistic structures and long-range contextual dependencies in input text, overcoming the limitations of traditional attention mechanisms for long-sequence processing.
- We design Mamba-Dec, an autoregressive decoder that combines causal state space modeling with cross-modal attention to generate temporally coherent facial landmark sequences while maintaining precise alignment with textual semantics across extended time horizons.
- We establish through extensive experiments that Mamba-Lip outperforms existing Transformer-based methods in both generation quality and efficiency, particularly for long sequences, while demonstrating robust performance across diverse linguistic inputs and speaking styles.

II. Related Work

A. Lip Reading Generation

Lip reading, as a crucial visual language, continues to receive significant research attention. Early studies primarily center on lip recognition, employing pixel-based or shape based methods that utilize lip pixel values or geometric features for classification. For instance, some works extract local image patches [5], combine geometric and PCA features [6], or use ASM models to generate features from key point coordinates [7]. Similar to sign language translation and production [8]–[13], lip language generation essentially constitutes the inverse process of recognition, which generally falls into audio-driven and landmark-based approaches. Audio-driven methods typically synthesize lip movements directly from speech signals. Early approaches combine 3D CNN and 2D-CNN for visual feature extraction [14]. More contemporary work includes Wav2Lip [1], which introduces a lip-sync discriminator to enhance synchronization accuracy. Subsequent research [15] uses a single reference image with implicit pose codes for identity preservation. Other studies incorporate visual mnemonics [2] or model facial dynamics through keypoint motion [16]. Landmark-based methods often use facial landmarks as intermediate representations. Some systems employ RNNs to synthesize mouth textures from audio [17], while others estimate facial landmarks from audio before generating video frames [18]. Methods like MakeItTalk [3] leverage self-attention and LSTMs for lip shape prediction, and GAN-based models handle artistic speaking video generation [19]. With advances in 3D face modeling, recent works also reconstruct 3D face meshes (3DMM) [20] from audio, enabling high-quality talking face generation by modeling expression parameters [21], [22].

B. Selective State Space Models

State Space Models (SSMs) [23]–[25] demonstrate strong capabilities in modeling long-range dependencies within sequential data. Notable models such as Mamba [4] and Mamba 2 [26] are now widely applied across various domains, including medicine [27], [28], object detection [29], [30], image processing [31], [32], natural language processing [33], [34], and time series analysis [35], [36]. The seminal S4 model [24] addresses long-range dependency issues through Hippo initialization [37], and its extension S4ND [38] further generalizes SSMs to multidimensional signals, enabling effective processing of large-scale visual data. Given the Transformer's [39]–[43] dominance in natural language modeling, recent efforts increasingly integrate Mamba with Transformer architectures to overcome the latter's limitations on long sequences. For example, Jamba [33] combines both in a hybrid architecture, achieving high throughput and low memory usage while maintaining competitive performance. E-Tamba [44] fine-tunes pre-trained Transformer and SSM models, showing strong results across NLP tasks. Zhang et al. [34] apply bidirectional Mamba to speech processing as an effective Transformer alternative. Other hybrid models, such as Mamba2-Hybrid [45], deliver faster inference without losing accuracy, and MambaVision [46] augments Mamba with self-attention modules to capture long-range spatial dependencies, excelling in vision tasks like detection and segmentation. Wang et al. [47] also explore multimodal fusion with visual Mamba, finding that incorporating attribute labels does not consistently improve performance.

III. Method

A. Overall Pipeline

In this work, we propose a novel Mamba-based framework for Lip Reading Generation, as illustrated in Figure 1. Given a text sentence $T = \{t_1, t_2, \cdots, t_n\}$ with N words, our lip generation system is required to generate a corresponding landmark sequence $L = \{l_1, l_2, \cdots, l_m\}$ with M frames. Additionally, we take ground-truth landmarks $\hat{L} = \{\hat{l}_1, \hat{l}_2, \cdots, \hat{l}_m\}$ as fitting targets during training.

B. Mamba-Enc: Modeling Linguistic Structure with SSMs

To capture the semantic features of input text sequences, we construct a Mamba-based encoder that effectively models long-range linguistic dependencies. Unlike traditional attention mechanisms, our approach employs selective state space models (SSMs) to process textual information in a recurrent yet parallelizable manner. First, a linear embedding layer transforms the discrete text tokens into continuous vector representations. We further enhance these embeddings with positional encoding to incorporate temporal order information. The embedding process is formulated as:

$$t'_n = W^t \cdot t_n + b^t + PE(n), \tag{1}$$

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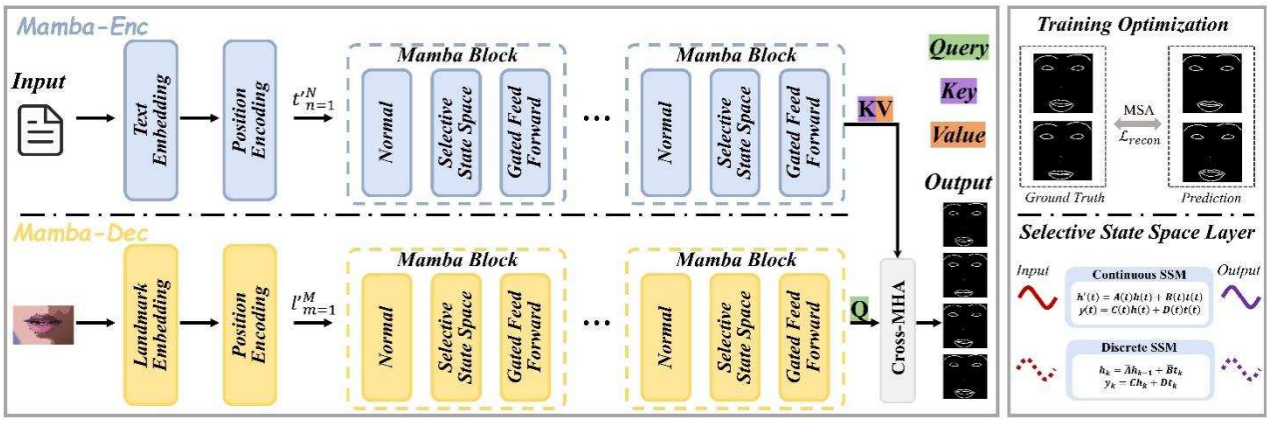


Figure 1. Overview of the proposed **Mamba-Lip**. The framework comprises two core components: (1) **Mamba-Encoder** (top-left), which transforms textual input into latent representations through stacked Mamba Blocks (each integrating Normal, Selective State Space, and Gated Feed Forward layers) after text embedding and positional encoding; (2) **Mamba-Decoder** (bottom-left), which processes landmark input via Landmark Embedding and positional encoding, then refines features through stacked Mamba Blocks and Cross Multi-Head Attention (Cross-MHA) for cross-modal feature fusion.

where t_n denotes the one-hot vector of the n-th token from the text vocabulary V, PE represents the sinusoidal positional encoding as introduced in [39], and W^t and b^t are learnable parameters of the embedding layer.

The resulting text embeddings $t'^N_{n=1}$ are then fed into the Mamba encoder to extract hierarchical semantic representa tions. The encoder comprises M identical Mamba blocks, each containing a Selective State Space Layer (SL), two Normalization Layers (NL), and a gated Feed forward Layer (GFL). The computational flow is defined as:

$$\begin{aligned} \{\tilde{t}\}^N_{n=1} &= TextEncoder(t'_{1:N}) \\ \Leftrightarrow \begin{cases} g_0 = \{t'_1, t'_2, \cdots, t'_N\}; \\ g_i = MambaBlock(g_{i-1}), i \in [1, X]; \\ \{\tilde{t}\}^N_{n=1} = g_X, \end{cases} \end{aligned} \tag{2}$$

where g_i represents the intermediate textual features after the i-th Mamba block. Each Mamba block operates as:

$$Block(h) = GFL(NL(SL(NL(h)))) \tag{3}$$

The core component of our encoder is the Selective State Space Layer (SL), which enables content-dependent reasoning over the input sequence. The SSM formulation is defined by the following continuous system:

$$\begin{aligned} h'(t) &= A(t)h(t) + B(t)t(t); \\ y(t) &= C(t)h(t) + D(t)t(t), \end{aligned} \tag{4}$$

where the system parameters A(t), B(t), C(t), and D(t) are dynamically generated based on the input tokens, allowing the model to selectively propagate or forget information throughout the sequence.

The selective mechanism enables the model to dynamically adjust its state transitions based on the semantic content of the input text, allowing it to focus on relevant linguistic contexts while ignoring redundant information.

C. Mamba-Dec: Autoregressive Visual Sequence Prediction

In this work, we aim to generate fine-grained facial land mark sequences. The landmark data for each time stamp contains 68 facial points, resulting in a 2D coordinate dimension. Similar to the text encoding, we encode the input landmarks into a high-dimensional feature space through a linear layer and a positional encoding layer PE'. The computational formulation is as follows:

$$l'_m = W^l \cdot l_m + b^l + PE'(m), \tag{5}$$

where l_m denotes the coordinates of the landmark at m-th time stamp; W^l and b^l represent the learnable weight and bias respectively.

Our landmark decoder is built on a recurrent Mamba architecture that autoregressively predicts the next landmark $\tilde{l}_{m+1}$ by sequentially processing all previously generated landmarks $\tilde{l}_{1:m}$ while leveraging a cross-modal attention mechanism combined with semantic guidance from encoded text features. The landmark decoding process can be formally expressed as:

$$\begin{aligned} \tilde{l}_{m+1} &= LandmarkDecoder(l'_{1:m}, \tilde{t}_{1:N}) \\ \Leftrightarrow \begin{cases} z_m = MambaBlock(l'_{1:m}) + \tilde{l}_m), m \in [1, M]; \\ \tilde{l}_{m+1} = Cross - MHA(z_m, \tilde{t}_{1:N}), \end{cases} \end{aligned} \tag{6}$$

where z_m denotes the intermediate landmark representations after the Mamba block. Here, $MambaBlock$ operates as a causal self-processing module that captures temporal dynamics within the landmark sequence, while $Cross - MHA$ facilitates semantic interaction between landmark features and encoded text representations.

Landmark Optimization: After M time stamps, we obtain the complete landmark sequence $\tilde{l}^M_{m=1}$ During training, we employ reconstruction loss to ensure both accuracy and temporal coherence of the generated landmarks. The recon struction loss is formulated as Mean Absolute Error (MSE) between predicted landmarks $\tilde{L} = \tilde{l}^M_{m=1}$ and ground truth $\hat{L} = \hat{l}^M_{m=1}$:

$$L_{recon} = \frac{1}{M} \sum^M_{m=1} |\tilde{l}_m - \hat{l}_m|. \tag{7}$$

IV. Experiments

A. Experimental Settings

Dataset. Following existing works [1], [50], [51], we experiment on a popular benchmark dataset GRID to evaluate the effectiveness of the proposed method. The

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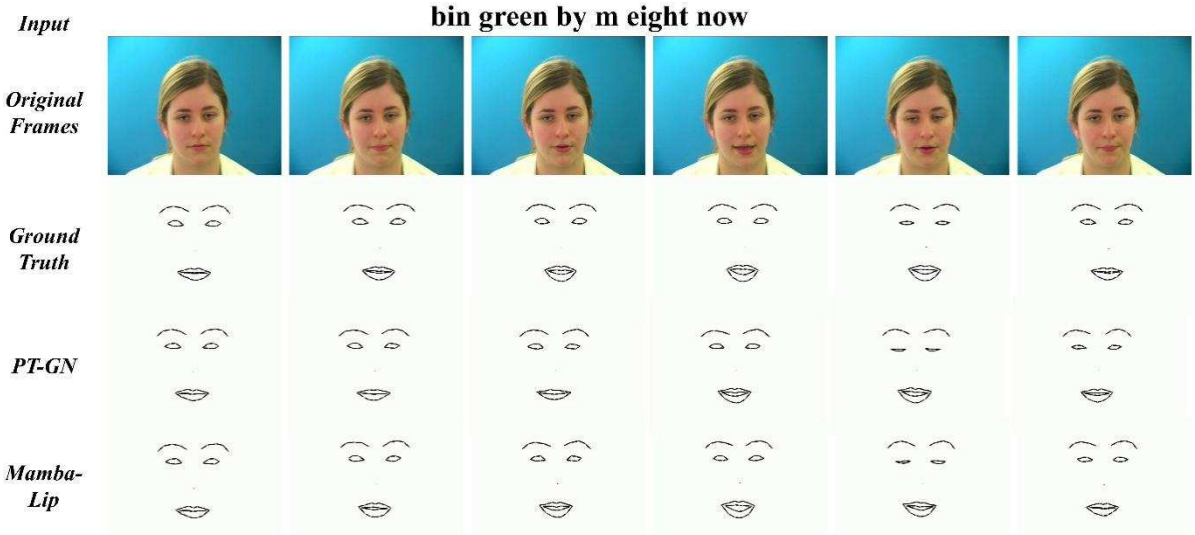


Figure 2. Visualization examples on GRID dataset.

TABLE I. QUANTITATIVE RESULTS ON GRID DATASET

Methods	DEV						TEST					
	BLEU1↑	BLEU2↑	BLEU3↑	BLEU4↑	ROUGE↑	WER↓	BLEU1↑	BLEU2↑	BLEU3↑	BLEU4↑	ROUGE↑	WER↓
PT-base [48]	29.08	15.82	9.06	3.12	29.88	67.15	29.32	14.97	9.21	3.55	30.01	68.37
GEN-OBT [10]	35.69	22.27	12.76	6.64	35.69	57.17	36.36	22.75	13.01	6.35	36.36	56.33
PT-GN [48]	40.97	28.32	17.60	10.03	40.97	51.90	40.87	28.26	17.34	9.76	40.87	52.17
LVMCN [49]	47.06	33.68	22.10	13.98	47.06	45.86	47.85	34.67	23.41	15.36	47.85	45.41
Mamba-Lip (Ours)	**48.21**	**35.71**	**24.72**	**15.57**	**48.21**	**44.92**	**48.56**	**36.03**	**24.54**	**15.84**	**48.13**	**44.68**

GRID dataset which is a widely used in lipreading has 34 speakers, but only 33 speakers' videos are available. There are 51 words in total, including 4 commands, 4 colors, 4 prepositions, 25 letters, 10 numbers and 4 adverbs where each sentence consists of a verb + color + preposition + letter + number + adverb (e.g., lay red with y two again). All videos are the same length and have a frame rate of 25fps.

Evaluation metrics. Since lip-speak and sign language, as special visual languages, have the same semantic features when they are generated, we adopt NSLT [52] for calculating metrics such as BLEU, ROUGE, and Word Error Rate (WER), which is widely used in sign language tasks [10], [48], [53], as an offline back-translation evaluation tool.

Data prepossessing. Since the CRID dataset does not provide pre-defined facial landmarks, we generate the ground-truth sequences through a standardized preprocessing pipeline. For each video frame, we first employ the DLib toolkit to detect the face and extract 68 facial landmarks. We subsequently focus on the 20 key points encompassing the inner and outer lip contours, which are most critical for articulatory representation. These 2D coordinates are then normalized to compensate for global head movements and scaled into a uniform range, resulting in the final landmark sequences used for model training and evaluation. In addition, we retrained the reverse translation model NSLT [52] on GRID separately under a fair setting for evaluating the effectiveness of our model in the following experiments.

Model settings. We build our method Mamba-Lip with the embedding size of 512 consistently. We apply a Gaussian noise onto pose coordinates in the pose embedding phase and the noise rate is set to 5. We train all parts of our network with the learning rate of $1*10^{-3}$ and batchsize is 64 with Adam optimizer. Our code implemented PyTorch on NVIDIA GeForce RTX 2080 Ti GPU.

B. Comparison with State-of-the-Arts

As shown in Table I, we conducted a comprehensive comparison with several state-of-the-art Transformer-based methods on the GRID dataset. Our proposed Mamba-Lip framework demonstrated superior performance across nearly all evaluation metrics, setting a new technical standard for lip sync generation. Specifically, Mamba-Lip achieved the highest BLEU score and lowest word error rate on both the development and test sets, significantly outperforming the previous best-performing Transformer-based method, LVMCN. These consistent performance improvements on both the development and test sets demonstrate the exceptional ability of selective state-space models to capture long-range dependencies in visual sequence generation. Notably, our method performed particularly well on high-level n-gram metrics, demonstrating its significant advantage in generating coherent and linguistically accurate landmark sequences. This breakthrough is attributed to the Mamba architecture's linear computational complexity and selective mechanism, which more effectively models temporal dynamics than traditional attention-based architectures. Experimental results demonstrate the significant advantages of selective state-space models for cross-modal sequence generation tasks, providing a new technical path for future research.

As illustrated in Figure 2, we present a comprehensive visual comparison of lip generation results for the input

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sequence "bin green by m eight now". The visualization clearly demonstrates the superior performance of our Mamba-Lip framework in generating accurate and natural lip movements. Compared to the PT-GN baseline, our method produces lip shapes that are remarkably closer to the Ground Truth across all frames. The generated lip landmarks maintain consistent temporal coherence and exhibit more natural deformation patterns throughout the entire utterance. These visual observations align well with our quantitative findings, further validating Mamba-Lip's enhanced capability in modeling the complex dynamics of lip movements for accurate visual speech generation.

TABLE II. NECESSITY OF A FULLY MAMBA-BASED ARCHITECTURE

Methods	BLEU1↑	BLEU2↑	BLEU3↑	BLEU4↑	ROUGE↑	WER↓
AM	32.56	19.95	11.12	5.42	32.56	67.26
MA	34.40	20.13	11.23	5.69	34.40	65.64
MM (Ours)	**48.56**	**36.03**	**24.54**	**15.84**	**48.13**	**44.68**

C. Ablation Study

As shown in Table II, we conducted ablation experiments to further explore the performance impact of different Mamba architecture configurations in both the encoder and decoder. Experimental results demonstrate that our proposed full Mamba architecture (**MM**) significantly outperforms variants using only portions of Mamba across all evaluation metrics, demonstrating the necessity of a fully Mamba-based design for lip sync generation.

Specifically, the **AM** configuration (using Mamba only in the decoder) and the **MA** configuration (using Mamba only in the encoder) perform comparably across all metrics, but exhibit significant differences compared to the full MM architecture. MM achieves significant improvements of 12.42 and 12.15 percentage points on the BLEU-4 metric compared to AM and MA, respectively, while also reducing the WER from approximately 66% to 44.68%, a relative improvement of over 30%. The results reveal the synergistic effect of Mamba in both the encoder and decoder: Mamba in the encoder effectively captures long-term semantic dependencies in text sequences, while Mamba in the decoder excels at modeling the temporal dynamics of visual sequences. The combination of the two forms a highly efficient cross-modal generation system.

V. CONCLUSIONS

In this paper, we proposed Mamba-Lip, a novel framework that revolutionizes lip reading generation by leveraging selective state space models. Our approach, built upon a fully Mamba-based encoder-decoder architecture, demonstrates superior capability in capturing long-range dependencies for cross-modal sequence generation. Extensive experiments on the GRID dataset show that Mamba-Lip achieves state-of-the-art performance, significantly outperforming existing Transformer-based methods. Ablation studies further validate the effectiveness of Mamba-2 and the necessity of our full Mamba architecture. This work establishes SSMs as a powerful paradigm for visual sequence generation and opens new avenues for efficient cross-modal synthesis.

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